

Question Number	Answer	Additional Guidance	Mark
3ai	An answer with the following : • metaphase and anaphase		1

Question Number	Answer	Additional Guidance	Mark
3aii	An answer which includes the following points: Similarities • cytokinesis (occurs) (1) • two (genetically) identical cells formed (1) Differences • cell plate formed in plants and not in animals / animal cell contracts but plant cell forms {vesicles / wall} in middle / plants cells remain connected by {plasmodesmata / pits}, none in animal cells. (1)	accept description, if implied under differences	3

Question Number	Answer	Additional Guidance			Mark									
3biii	A table with the following features : • suitable table drawn (1) • all headings correct (1) • all data correctly entered (1)	<table><tr><td></td><td>number of cells in Interphase</td><td>number of cells in a stage of Mitosis</td></tr><tr><td>Control</td><td>148</td><td>24</td></tr><tr><td>Treated with lectin</td><td>160</td><td>88</td></tr></table>		number of cells in Interphase	number of cells in a stage of Mitosis	Control	148	24	Treated with lectin	160	88	table must have 2 rows and 2 columns minimum Do not allow .0 Extras lose mp 3		3
	number of cells in Interphase	number of cells in a stage of Mitosis												
Control	148	24												
Treated with lectin	160	88												

Question Number	Answer	Additional Guidance	Mark
3biv	A calculation that includes the following steps : • determine total number of cells in mitosis in lectin treatment and total number of cells observed (1) • calculate percentage (1)	e.g. 88 and 160+88=248 $(88/248) \times 100 = 35.5 / 36$	2

Question Number	Answer	Additional Guidance	Mark
3bv	An answer that includes the following points : - percentage in mitosis is greater with lectins than without (1) - but we have no idea of variability / difference may not be significant / no SDs (1)	IGNORE number of cells	2

Question Number	Answer	Additional Guidance	Mark
3bvi	An answer that includes the following points : - use of Chi (squared test) (1) - because need to test {the significance of the difference between observed and expected results / association / frequency data} (1)	Allow ecf for t-test	2